

THE LEVEL-SHIFT LOGIC OF A CATEGORY: MODAL CORRESPONDENCE FOR SAMENESS LATTICES, QUOTIENT MODELS WITH CONTINGENT IDENTITY, AND COMPLETENESS

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ABSTRACT. An object is known only up to a chosen notion of sameness. Taking this literally, we attach to a small category C its *level-shift frame* $F(C) = (\text{Same}(C), \subseteq)$: the worlds are the weak-equivalence structures on C (classes of morphisms containing identities, closed under composition, satisfying two-out-of-three), ordered by coarsening, and $\Box\varphi$ reads “ φ at every coarser level”. We determine which modal axioms this frame validates and what they say about C . Every $F(C)$ validates S4.2; it validates S4.3 iff $\text{Same}(C)$ is a chain, and for a finite group G (where $\text{Same}(G)$ is the subgroup lattice) iff G is a cyclic p -group. The bounded-width and bounded-depth axioms bw_n , bd_n correspond exactly to $\text{width Same}(C) \leq n$ and $\text{height Same}(C) \leq n$, so every finite bound on the width and height of a subgroup lattice is modally detectable; by contrast, distributivity and modularity of $\text{Same}(C)$ are provably not modally definable over sameness frames, and already between group frames: the Boolean $\text{Same}(C_{30})$ maps p-morphically onto $\text{Same}(V_4) \cong M_3$, and the Fano-plane lattice $\text{Same}(C_2^3)$ onto the non-modular $\text{Same}(A_4)$. The intended models are diagrams of quotients $W \mapsto \text{Ob}(C)/\sim_W$ with surjective transitions, i.e. counterpart-style modal models with contingent, monotone identity; the identity/encapsulation fragment of the resulting logic is sound and complete for abstract such models (with computational validation of the canonical construction), and completeness for the category-generated models is controlled by a realization/separation problem for Same. Finally we show that every finite bounded poset with at most six elements is a p-morphic image of a finite sameness frame, so the logic of finite sameness frames agrees with S4.2Grz on all frames of that size and on all finite chains; the logic of all sameness frames is strictly weaker, as infinite ones violate Grz. All finite claims are verified by exhaustive computation, with explicit p-morphism certificates supplied as supplementary material.

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1. INTRODUCTION

In an object-oriented reading of mathematics an object is never given absolutely: it is observed through an *interface*, and two objects that the interface cannot distinguish are, for the purposes of that interface, the same. Homotopy theory, categorical logic and the theory of systems each have a name for this — weak equivalence, the Structure Identity Principle, behavioural equivalence — and in each case “the same” is a *choice*: a class of morphisms declared to be identifications.

The companion paper [1] organises all such choices on a small category C into a single complete lattice, the *sameness lattice* $\text{Same}(C)$, and develops it as a categorical invariant (it is the subgroup lattice on a group, and a functor into Galois connections). The present paper asks a different, logical question:

If a level of sameness is a world, and moving to a coarser level is accessibility, what is the modal logic of a category — and what does it say about the category?

We call the resulting frame $F(C) = (\text{Same}(C), \subseteq)$ the *level-shift frame* and its modal logic the *level-shift logic* of C . Since $\subseteq$ is a preorder, the logic contains S4; the interest is in what lies above S4 and in how the extra axioms read off the structure of C .

Contributions.

- (1) *Axioms detect shape* (§3). $F(C)$ validates S4.2 for every C ; it validates S4.3 iff $\text{Same}(C)$ is a chain; for a finite group G this holds iff G is a cyclic p -group, with C_6 the sharp counterexample. Finite C give Grz, infinite C need not.
- (2) *Correspondence and its limit* (§4). The bounded-width and bounded-depth axioms correspond to the width and height of $\text{Same}(C)$; for a group, every finite bound on the width and height of its subgroup lattice is thereby modally definable. Distributivity and modularity are *not* modally definable over sameness frames, witnessed by p -morphisms between group frames: $\text{Same}(C_{30}) \twoheadrightarrow \text{Same}(V_4)$ and $\text{Same}(C_2^3) \twoheadrightarrow \text{Same}(A_4)$.
- (3) *Models and proof theory* (§5). The intended semantics is a diagram of quotients with surjective transitions — a counterpart-style modal semantics with contingent, monotone identity. With an explicit language (levels, individuals, observable symbols, value equality) the identity/encapsulation fragment is sound and complete for abstract such models by an explicit canonical-model argument, validated computationally; completeness for the category-generated models is controlled by a realization/separation problem for Same .
- (4) *Completeness up to six elements* (§6). Every finite bounded poset with ≤ 6 elements — including one that is not a lattice — is a p -morphic image of a finite sameness frame; every finite chain is a

sameness frame. Hence the finite level-shift logic agrees with S4.2Grz on all such frames; the full equality is left open.

Every finite assertion is verified by exhaustive computation; the scripts are named inline (§7).

Related work. The axioms S4.2, S4.3, Grz and the bounded width/depth families $\mathbf{bw}_n$, $\mathbf{bd}_n$ and their frame conditions are classical [3, 4, 5]; we use them as a measuring instrument rather than study them for their own sake. The closest kin is van Benthem and Bezhanishvili’s modal analysis of groups and vector spaces [2], which equips algebraic structures with modal languages and studies definability, axiomatisation and p-morphisms; our frames arise differently — as lattices of weak-equivalence structures, so that modal axioms become statements about a category and, via the identification $\text{Same}(G) = \text{Sub}(G)$ of [1], about the subgroup lattice of a group — but the programme is the same in spirit, and the results here can be read as modal invariants of groups in their sense. Contingent identity in first-order modal logic goes back to [6]; our models arrive at it from the opposite direction, as quotients that only ever collapse. The object-oriented reading connects to the Structure Identity Principle [12] and to behavioural equivalence in coalgebra [11]. Subgroup lattices as an invariant are the subject of [9]; Ore’s characterisation of distributivity [10] enters through Theorem 3.4.

2. PRELIMINARIES

2.1. Samenesses. A *sameness* on a small category C is a class $W \subseteq \text{Mor}(C)$ containing all identities, closed under composition, and satisfying two-out-of-three (if two of $f, g, g \circ f$ lie in W , so does the third). $\text{Same}(C)$ is the set of samenesses ordered by inclusion. The following basic facts are from [1]; we include the short proofs so that this paper is self-contained.

Proposition 2.1. *Same(C) is a complete lattice, with meet $\cap$, least element $\perp = \{\text{id}\}$ and greatest element $\top = \text{Mor}(C)$. For a group G viewed as a one-object category, $\text{Same}(G) = \text{Sub}(G)$, the subgroup lattice; for a groupoid, the lattice of wide subgroupoids. $\text{Same}(C_1 \amalg C_2) \cong \text{Same}(C_1) \times \text{Same}(C_2)$.*

Proof. Each of the three defining conditions is preserved by arbitrary intersections (if two of $f, g, g \circ f$ lie in every W_i , the third lies in each W_i), and $\text{Mor}(C)$ satisfies them; a meet-complete poset with a top is a complete lattice, and $\{\text{id}\}$ is a sameness contained in every other. For a group: a subgroup contains e , is closed under products and satisfies two-out-of-three (the third of f, g, gf is a product of the other two and their inverses). Conversely a sameness W contains e and is closed under products; for $w \in W$ the composable pair (w^{-1}, w) has composite $e \in W$ and factor $w \in W$, so two-out-of-three gives $w^{-1} \in W$; hence W is a subgroup. The same argument, applied at each object, gives the wide-subgroupoid statement. Finally, in a coproduct no morphism connects the two summands, so composites and

two-out-of-three instances stay within a summand, and a sameness of $C_1 \amalg C_2$ is exactly a pair of samenesses. $\square$

We write $\sim_W$ for the equivalence relation on $\text{Ob}(C)$ generated by zigzags of morphisms in W , and $\llbracket C \rrbracket_W = \text{Ob}(C)/\sim_W$; $W \subseteq W'$ gives a surjection $\llbracket C \rrbracket_W \twoheadrightarrow \llbracket C \rrbracket_{W'}$.

2.2. Modal frames. A (reflexive, transitive) frame (X, R) validates S4. We use the following standard axioms and their first-order frame conditions [4, 5]:

- .2 ($\Diamond \Box p \rightarrow \Box \Diamond p$): *directedness*, $wRu \wedge wRv \Rightarrow \exists x (uRx \wedge vRx)$;
- .3 ($\Box(\Box p \rightarrow q) \vee \Box(\Box q \rightarrow p)$): *weak connectedness*, $wRu \wedge wRv \Rightarrow uRv \vee vRu$;
- bw_n ($\bigwedge_{i \leq n} \Diamond p_i \rightarrow \bigvee_{i \neq j} \Diamond(p_i \wedge \Diamond p_j)$): no $(n+1)$ -antichain inside any R -successor set $R[w]$; bw_1 is .3;
- bd_n : no strictly ascending R -chain of $n+1$ points from any w ;
- Grz ($\Box(\Box(p \rightarrow \Box p) \rightarrow p) \rightarrow p$): on finite frames, R is a partial order.

A *p-morphism* $f: (X, R) \rightarrow (Y, S)$ is a map with $wRu \Rightarrow f(w)Sf(u)$ and $f(w)Sv \Rightarrow \exists u (wRu \wedge f(u) = v)$; a *generated subframe* is an R -upward-closed subset. Modal validity is preserved under p-morphic images and generated subframes, and a finite rooted frame is determined up to these operations by its Jankov–Fine formula [4]. The logic of a class $\mathcal{K}$ of frames is $\text{Log}\mathcal{K} = \{\varphi : \mathcal{K} \models \varphi\}$.

3. THE LEVEL-SHIFT FRAME AND ITS AXIOMS

Definition 3.1. The *level-shift frame* of C is $\text{F}(C) = (\text{Same}(C), \subseteq)$; the *level-shift logic* of C is $\text{Log}\{\text{F}(C)\}$. We distinguish

$$\text{Log}_{\text{all}} := \text{Log}\{\text{F}(C) : C \text{ small}\}, \quad \text{Log}_{\text{fin}} := \text{Log}\{\text{F}(C) : C \text{ finite}\},$$

so that $\text{Log}_{\text{all}} \subseteq \text{Log}_{\text{fin}}$.

Since $\subseteq$ is a preorder, $\text{F}(C) \models \text{S4}$: $\Box\varphi$ holds at level W iff φ holds at every coarser level $W' \supseteq W$.

Theorem 3.2. *For every small category C , $\text{F}(C) \models \text{S4.2}$.*

Proof. Directedness: given $W \subseteq U$ and $W \subseteq V$, take $X = U \vee V$, which exists since $\text{Same}(C)$ is a complete lattice (Proposition 2.1); $\top$ would also do. $\square$

Theorem 3.3. *$\text{F}(C) \models \text{S4.3}$ iff $\text{Same}(C)$ is a chain.*

Proof. If $\text{Same}(C)$ is a chain, any two successors of a world are comparable. Conversely, put $W = \perp = \{\text{id}\}$, the least sameness: every U, V are successors of $\perp$, so weak connectedness at $\perp$ makes every pair comparable. $\square$

Theorem 3.4. *For a finite group G , $\text{F}(G) \models \text{S4.3}$ iff G is a cyclic p -group.*

Proof. By Proposition 2.1, $\text{Same}(G) = \text{Sub}(G)$, so by Theorem 3.3 the question is when $\text{Sub}(G)$ is a chain. If two primes $p \neq q$ divide $|G|$, Cauchy's theorem gives subgroups of orders p and q , which are incomparable. So G is a p -group; a finite p -group whose subgroup lattice is a chain has a unique maximal subgroup, hence is cyclic (Burnside's basis theorem: $G/\Phi(G)$ is elementary abelian of rank one), and the subgroup lattice of a cyclic p -group is a chain [9]. $\square$

Remark 3.5 (infinite groups). Finiteness is essential: the Prüfer group C_{p^∞} has subgroup lattice $1 < C_p < C_{p^2} < \dots < C_{p^\infty}$, a chain, so $\mathbf{F}(C_{p^\infty}) \models \text{S4.3}$ although C_{p^∞} is not cyclic. The groups with totally ordered subgroup lattice are exactly the cyclic p -groups and the Prüfer p -groups [9]; with that classification Theorem 3.4 extends verbatim to all groups.

Example 3.6. C_6 is cyclic but not a p -group: $\text{Sub}(C_6) = \{1, C_2, C_3, C_6\}$ has the incomparable pair C_2, C_3 (Figure 1), so $\mathbf{F}(C_6) \not\models \text{S4.3}$ although $\mathbf{F}(C_6) \models \text{S4.2}$. By contrast $\text{Sub}(C_4)$, $\text{Sub}(C_8)$, $\text{Sub}(C_9)$ are chains.

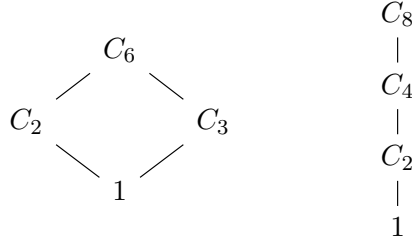


FIGURE 1. $\mathbf{F}(C_6)$ (left) refutes .3 at the root: C_2 and C_3 are incomparable successors of 1. $\mathbf{F}(C_8)$ (right) is a chain and validates S4.3.

Alt text: Two Hasse diagrams. Left, the four subgroups of C_6 : 1 at the bottom, C_2 and C_3 side by side above it and incomparable, C_6 at the top.

Right, the four subgroups of C_8 in a single vertical chain 1, C_2 , C_4 , C_8 .

Proposition 3.7. *For finite C , $\mathbf{F}(C) \models \text{Grz}$; thus $\text{Log}_{\text{fin}} \supseteq \text{S4.2Grz}$, and the level-shift logic of a finite C contains S4.3Grz exactly when $\text{Same}(C)$ is a chain.*

Proof. $\mathbf{F}(C)$ is a finite partial order, and finite partial orders validate Grz [4]. $\square$

Example 3.8 (Grz fails for infinite C). Let C have two objects x, y and countably many parallel arrows $f_1, f_2, \dots : x \rightarrow y$ with no non-identity composites. Every set of arrows is a sameness, so $\text{Same}(C) \cong \mathcal{P}(\mathbb{N})$, which contains the infinite strictly ascending chain $\{f_1\} \subset \{f_1, f_2\} \subset \dots$; hence $\mathbf{F}(C) \not\models \text{Grz}$ [4]. So $\text{Log}_{\text{all}} \not\supseteq \text{Grz}$, while $\text{Log}_{\text{all}} \supseteq \text{S4.2}$ by Theorem 3.2, which used only that $\text{Same}(C)$ is a complete lattice. The two logics of Definition 3.1 are therefore genuinely different, and §6 concerns Log_{fin} .

Theorems 3.2–3.4 and Example 3.6 are verified on $C_4, C_8, C_9, C_6, V_4, S_3, Q_8$ and two poset categories in `level_logic.py`.

4. CORRESPONDENCE THEORY AND ITS LIMIT

Theorem 4.1. *For every small C and $n \geq 1$: $F(C) \models \text{bw}_n$ iff $\text{width Same}(C) \leq n$, and $F(C) \models \text{bd}_n$ iff $\text{height Same}(C) \leq n$ (here $\text{width Same}(C) \leq n$ means that $\text{Same}(C)$ has no antichain of $n+1$ elements, and $\text{height Same}(C) \leq n$ that it has no chain of $n+1$ elements; for finite C these are the usual width and height).*

Proof. The frame conditions are local to successor sets $R[w] = \{U : W \subseteq U\}$. Since $\perp$ is the least sameness, $R[\perp] = \text{Same}(C)$, so the conditions at $\perp$ are the global bounds; conversely every $R[w] \subseteq \text{Same}(C)$ inherits them. $\square$

Corollary 4.2. *For a group G and every finite n , the properties $\text{width Sub}(G) \leq n$ and $\text{height Sub}(G) \leq n$ are modally definable, by $F(G) \models \text{bw}_n$, resp. bd_n ; for finite G the width and height themselves are the least such n . (Infinite widths or heights are not separated by the finitary bw_n, bd_n .)*

Table 1 records the data; e.g. $F(S_3)$ validates $\text{S4.2Grz} + \text{bw}_4 + \text{bd}_3$ and refutes bw_3 , and $F(Q_8)$ validates $\text{S4.2Grz} + \text{bw}_3 + \text{bd}_4$ and refutes bw_2 and bd_3 . (The exact logic of a single finite frame requires in addition its Jankov–Fine formula; see Remark 5.3.)

C	$ \text{Same}(C) $	chain	width	height	first $\text{bw}_n / \text{bd}_n$
C_4	3	yes	1	3	$\text{bw}_1 / \text{bd}_3$
C_6	4	no	2	3	$\text{bw}_2 / \text{bd}_3$
C_8	4	yes	1	4	$\text{bw}_1 / \text{bd}_4$
V_4	5	no	3	3	$\text{bw}_3 / \text{bd}_3$
S_3	6	no	4	3	$\text{bw}_4 / \text{bd}_3$
Q_8	6	no	3	4	$\text{bw}_3 / \text{bd}_4$
poset $0 < 1 < 2$ (M_3)	5	no	3	3	$\text{bw}_3 / \text{bd}_3$
poset $0 < 1 < 2 < 3$	15	no	7	4	$\text{bw}_7 / \text{bd}_4$

TABLE 1. Level-shift frames: chain-ness (.3), width (bw_n) and height (bd_n). Computed in `level_logic.py`, `level_logic2.py`; the general frame conditions were checked independently of the global invariants and agree in every case.

The order-shape of $\text{Same}(C)$ is thus modally visible. Its lattice-theoretic content is not. Since modal validity is preserved under p-morphic images, a property $\mathcal{P}$ of sameness frames is not definable by a modal formula (uniformly, over the class of sameness frames) as soon as there is a p-morphism $F(C) \rightarrow F(D)$ with $F(C) \in \mathcal{P}$ and $F(D) \notin \mathcal{P}$. Both ends must be sameness frames: an image outside the class proves nothing (for instance $F(C_{12})$, a

2×3 grid, maps p -morphically onto the pentagon N_5 , but N_5 is not known to be a sameness frame, cf. Lemma 6.3).

Theorem 4.3 (distributivity is not definable). *Let p, q, r be distinct primes and $G = C_{pqr}$ (for instance C_{30}). Then $\text{Same}(G) = \text{Sub}(G)$ is the divisor lattice of pqr , the Boolean lattice 2^3 with atoms C_p, C_q, C_r , which is distributive; $\text{Same}(V_4) \cong M_3$ is not. The map $f: \text{Sub}(C_{pqr}) \rightarrow \text{Sub}(V_4)$,*

$$1 \mapsto 1;$$

$$C_p, C_q, C_r \mapsto \text{the three subgroups of order 2 (bijectively)};$$

$$H \mapsto V_4 \text{ if } |H| \text{ is divisible by at least two of } p, q, r,$$

is a p -morphism. Hence no modal formula defines distributivity of $\text{Same}(C)$ over sameness frames.

Proof. Subgroups of a cyclic group correspond to divisors of its order; for square-free pqr the divisor lattice is 2^3 , hence distributive. f is order-preserving. Back condition: from 1 every element of M_3 is reached by some subgroup; from C_p the successors are its image and V_4 , reached by C_p and C_{pq} ; from a subgroup whose order is divisible by two of the primes the only successor V_4 is reached by itself. Surjective. Both frames are subgroup lattices, hence sameness frames (Proposition 2.1). $\square$

Remark 4.4. The same lattice and the same map arise from the category D_3 with two objects and three parallel arrows a, b, c and no non-identity composites: every set of arrows is a sameness, so $\text{Same}(D_3) \cong 2^{\{a, b, c\}}$. The group witness is chosen to match Theorem 4.5: both blind spots are then exhibited between subgroup lattices.

Theorem 4.5 (modularity is not definable). *$\text{Same}(C_2^3) = \text{Sub}(C_2^3)$ is the lattice of subspaces of $\mathbb{F}_2^3$ — bottom, the seven points and seven lines of the Fano plane, top — which is modular. $\text{Same}(A_4) = \text{Sub}(A_4)$ is not modular. Fix a line ℓ_0 of the Fano plane and define $f: \text{Sub}(C_2^3) \rightarrow \text{Sub}(A_4)$ by*

$$\begin{aligned} 1 &\mapsto 1; & \text{the three points on } \ell_0 &\mapsto \text{the three subgroups of order 2 (bijectively)}; \\ \ell_0 &\mapsto V_4; & \text{the four points off } \ell_0 &\mapsto \text{the four subgroups of order 3 (bijectively)}; \\ & & \text{every other line, and the top} &\mapsto A_4. \end{aligned}$$

Then f is a p -morphism. Hence no modal formula defines modularity of $\text{Same}(C)$ over sameness frames.

Proof. $\text{Sub}(A_4)$ consists of 1, the three subgroups $\langle (12)(34) \rangle, \langle (13)(24) \rangle, \langle (14)(23) \rangle$ of order 2, the unique V_4 containing them, the four subgroups of order 3, and A_4 ; it contains the pentagon $1 < \langle (12)(34) \rangle < V_4 < A_4, 1 < \langle (123) \rangle < A_4$, so it is not modular. The subspace lattice of $\mathbb{F}_2^3$ is modular, being a projective geometry. Forth: a point $p \leq \ell$; if $\ell = \ell_0$ then $p \in \ell_0$ and $f(p) \leq V_4 = f(\ell_0)$; otherwise $f(\ell) = A_4$. Back: for $f(p) =$ an order-2 subgroup, its successors V_4 and A_4 are reached by $\ell_0 \ni p$ and by either other line through p ; for $f(q) =$ an order-3 subgroup ($q \notin \ell_0$), the successor A_4 is reached by any line through q (none of which is ℓ_0); for $f(\ell_0) = V_4$ the

successor A_4 is reached by the top; the bottom reaches everything since f is surjective. Both frames are subgroup lattices, hence sameness frames. The map is recorded as a computationally verified certificate (§7). $\square$

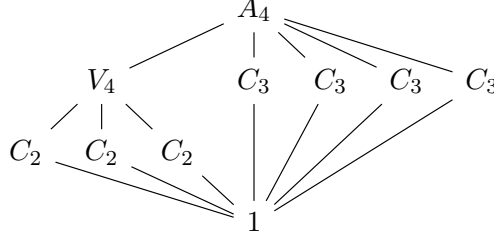


FIGURE 2. $\text{Sub}(A_4)$, the non-modular target of Theorem 4.5. Under f the three C_2 are the images of the points of a line ℓ_0 of the Fano plane, V_4 the image of ℓ_0 itself, the four C_3 the images of the points off ℓ_0 , and A_4 the image of the six other lines and the top.

Alt text: Hasse diagram of the ten subgroups of A_4 : the trivial group at the bottom; above it, on the left, three subgroups C_2 joined to a single V_4 above them, and on the right four subgroups C_3 ; V_4 and the four C_3 are all joined to A_4 at the top.

The dichotomy is clean: *the level-shift logic sees width, height, chain-ness and directedness of $\text{Same}(C)$, and provably cannot see its lattice identities.* (Each individual finite frame is of course pinned down by its Jankov–Fine formula; Theorems 4.3–4.5 are about a *uniform* axiom.) Both witnesses are group frames: the second says that the Fano plane with a distinguished line maps p-morphically onto the subgroup lattice of A_4 .

5. QUOTIENT MODELS AND THE LOGIC $\text{OO}(C)$

5.1. The intended models. The object-level semantics of [1] is a diagram (copresheaf) of quotients over the frame,

$$\text{Obs}_C: (\text{Same}(C), \subseteq) \rightarrow \mathbf{Set}, \quad W \mapsto \llbracket C \rrbracket_W = \text{Ob}(C)/\sim_W,$$

with surjective, functorial transition maps $\pi_{W \leq W'}: \llbracket C \rrbracket_W \rightarrow \llbracket C \rrbracket_{W'}$ (Figure 3). Read as a modal model of individuals: an *individual at level W* is an object up to $\sim_W$; a *W -observable* is a predicate on $\llbracket C \rrbracket_W$; passing to a coarser level sends every individual forward and may identify distinct ones. Hence identity is *contingent* — $a = b$ is world-dependent — and *monotone*: $a \sim_W b$ implies $a \sim_{W'} b$ for $W \subseteq W'$. This is a counterpart-style modal semantics with surjective transition maps, in the family of contingent-identity systems [6], with the special feature that identity only ever grows; we do not add quantifiers here, and treat the identity/observable layer as an atomic fragment.

$$\begin{array}{ccc}
\text{Ob}(C) & \xrightarrow{\pi_W} & \llbracket C \rrbracket_W \\
\pi_{W'} \downarrow & \swarrow \pi_{W \leq W'} & \\
\llbracket C \rrbracket_{W'} & &
\end{array}$$

FIGURE 3. The quotient model Obs_C : domains collapse along accessibility. A W -observable factors through π_W ; a W' -observable is a W -observable by pullback along $\pi_{W \leq W'}$.

Alt text: A commutative triangle. The set of objects of C at the top left maps by a surjection to the quotient at level W on the right and by a surjection to the quotient at level W' below; a third surjection from the level- W quotient to the level- W' quotient makes the triangle commute.

5.2. Language, abstract models, and the identity/encapsulation fragment. Fix a finite lattice L of *levels*, a set X_0 of *individual constants*, and a set Φ of *observable symbols*. The *atomic judgements* are

$$a \equiv_W b \ (W \in L), \quad \varphi \text{ obs}_W \ (\varphi \in \Phi, \ W \in L), \quad \varphi(a) \doteq \varphi(b) \ (\varphi \in \Phi),$$

read “ a, b are the same at level W ”, “ φ is observable at level W ”, “ φ takes the same value at a and b ”. An *abstract model* is $M = (X, (E_W)_{W \in L}, (\varphi^M)_{\varphi \in \Phi})$ with $X \supseteq X_0$ a set (constants denoting themselves), each E_W an equivalence relation on X with $E_W \subseteq E_{W'}$ whenever $W \leq W'$, and each $\varphi^M: X \rightarrow V_\varphi$ a function to some value set. Truth: $M \models a \equiv_W b$ iff $(a, b) \in E_W$; $M \models \varphi(a) \doteq \varphi(b)$ iff $\varphi^M(a) = \varphi^M(b)$; $M \models \varphi \text{ obs}_W$ iff φ^M is constant on every E_W -class. Every Obs_C (with $L = \text{Same}(C)$, $E_W = \sim_W$, and any functions on $\text{Ob}(C)$ as observables) is an abstract model. Over a set Γ of atomic assumptions the rules are

- (E) $\equiv_W$ is reflexive, symmetric and transitive, for each W ;
- (V) $\varphi(\cdot) \doteq \varphi(\cdot)$ is reflexive, symmetric and transitive, for each φ ;
- (coarsen) $a \equiv_W b, \ W \leq W' \Rightarrow a \equiv_{W'} b$;
- (restrict) $\varphi \text{ obs}_{W'}, \ W \leq W' \Rightarrow \varphi \text{ obs}_W$;
- (encapsulate) $\varphi \text{ obs}_W, \ a \equiv_W b \Rightarrow \varphi(a) \doteq \varphi(b)$.

$\Gamma \vdash \gamma$ means γ is in the closure of Γ under these rules.

Theorem 5.1. *For atomic Γ and atomic γ : $\Gamma \vdash \gamma$ iff $M \models \gamma$ for every abstract model $M \models \Gamma$.*

Proof. Soundness. (E), (V) hold because E_W and the kernel of φ^M are equivalence relations; (coarsen) by monotonicity of (E_W) ; (restrict) because a function constant on the classes of $E_{W'}$ is constant on the finer classes of $E_W \subseteq E_{W'}$; (encapsulate) by the definition of $\varphi \text{ obs}_W$.

Completeness. Write D for the closure of Γ . *Canonical model* M_0 : $X = X_0$; $E_W := \{(a, b) : a \equiv_W b \in D\}$, an equivalence relation by (E) and monotone by (coarsen); for each φ let $\doteq_\varphi := \{(a, b) : \varphi(a) \doteq \varphi(b) \in D\}$,

an equivalence relation by (V), and put $V_\varphi := X/\dot{=}_\varphi$ and $\varphi^{M_0}(a) := [a]$. Then $M_0 \models a \equiv_W b$ iff $a \equiv_W b \in D$, and $M_0 \models \varphi(a) \dot{=} \varphi(b)$ iff $\varphi(a) \dot{=} \varphi(b) \in D$, by construction; and $M_0 \models \Gamma$: its $\equiv$ - and $\dot{=}$ -atoms are in D , and if $\varphi \text{ obs}_W \in \Gamma$ then for $(a, b) \in E_W$ (encapsulate) gives $\varphi(a) \dot{=} \varphi(b) \in D$, i.e. $[a] = [b]$, so φ^{M_0} is constant on E_W -classes. Hence a non-derivable $a \equiv_W b$ or $\varphi(a) \dot{=} \varphi(b)$ is refuted by M_0 .

For a non-derivable $\varphi_0 \text{ obs}_{W_0}$ the model M_0 may still satisfy it (e.g. when $\Gamma = \emptyset$, all E_W are discrete), so we enlarge it. *Model M_1* : $X = X_0 \cup \{x, y\}$ with x, y fresh; $E'_W := E_W \cup \{(x, x), (y, y)\} \cup \{(x, y), (y, x)\}$ if $W \geq W_0$, and $E'_W := E_W \cup \{(x, x), (y, y)\}$ otherwise — equivalence relations, monotone because $W \geq W_0$ is upward closed; $\varphi_0^{M_1}(x), \varphi_0^{M_1}(y)$ two fresh distinct values, and $\psi^{M_1}(x) = \psi^{M_1}(y)$ a fresh value for $\psi \neq \varphi_0$; on X_0 all functions are as in M_0 . Then $M_1 \not\models \varphi_0 \text{ obs}_{W_0}$, since $(x, y) \in E'_{W_0}$ carry different values. And $M_1 \models \Gamma$: atoms of Γ mention only X_0 and hold as in M_0 ; for $\psi \text{ obs}_W \in \Gamma$ with $\psi \neq \varphi_0$ the class $\{x, y\}$ (if present) carries equal values; and if $\varphi_0 \text{ obs}_W \in \Gamma$ then $W \not\geq W_0$ — otherwise (restrict) would derive $\varphi_0 \text{ obs}_{W_0}$ — so x, y are not E'_W -related and φ_0 is constant on the singleton classes $\{x\}, \{y\}$. $\square$

The construction has been validated computationally: over the level lattices 3-chain, 2^2 and M_3 , three constants and two observable symbols, for 723 randomly generated Γ and every non-derivable atomic goal (22,413 goals), the model M_0 or M_1 satisfies Γ and refutes the goal (`oo_completeness.py`).

Remark 5.2 (relation to realization). Theorem 5.1 is completeness for abstract models. For the category-generated models $\{\text{Obs}_C\}$ less is needed than representing every abstract model: it suffices that every non-derivable pair (Γ, γ) be refuted by *some* Obs_C (a separation property). A sufficient route is a representation theorem: if every finite abstract countermodel — such as M_0 and M_1 above — can be embedded into, or otherwise faithfully represented by, some Obs_C with $\text{Same}(C) \cong L$, preserving the levels, the equivalence relations and the observables, then completeness transfers to the category-generated semantics. Whether such representations exist is a *realization problem* for Same: which lattices L , and which monotone families of equivalence relations with observables, arise from a category. §6 shows the lattice part is non-trivial already at five elements. Thus completeness for the category-generated semantics is controlled by a realization/separation problem for Same, which we leave open.

Remark 5.3 (propositional fragment; cut-elimination). For fixed finite C the propositional level-shift fragment is the logic of the single finite frame $F(C)$: tabular, decidable, with the finite model property, and axiomatised over S4.2Grz by the appropriate bw_n , bd_n and Jankov–Fine formulas [4]. Cut-free presentations are inherited: S4.2 has cut-free hypersequent calculi [7] and Grz a cut-free sequent calculus [8]. We do *not* claim cut-elimination for the combined system (modality + identity + observables); the fragment of Theorem 5.1 is Horn-style, its only cut being transitivity.

6. COMPLETENESS OF S4.2Grz UP TO SIX ELEMENTS

By Theorem 3.2 and Proposition 3.7, $\text{Log}_{\text{fin}} \supseteq \text{S4.2Grz}$ (Example 3.8 shows this fails for Log_{all}). Equality would follow if every finite rooted S4.2Grz-frame were a p-morphic image of a generated subframe of some $\mathbf{F}(C)$ with C finite. Finite rooted S4.2Grz-frames are exactly the finite *bounded posets*: Grz makes a finite frame a partial order, the root is its bottom, and directedness with finiteness gives a top [4]. So the question is which finite bounded posets are p-morphic images of sameness frames. Note that this is weaker than being *isomorphic* to one, and the two questions have different answers.

Theorem 6.1 (chains). *Every finite chain is a sameness frame: the $(n+1)$ -chain is $\mathbf{F}(C_{p^n})$.*

Proof. $\text{Same}(C_{p^n}) = \text{Sub}(C_{p^n})$ is the chain of subgroups of orders $1, p, \dots, p^n$ (Theorem 3.4). $\square$

Proposition 6.2 (five elements). *The five bounded posets with five elements are the 5-chain, M_3 , N_5 , and the two “square-plus-a-point” lattices $1 \oplus 2^2$, $2^2 \oplus 1$. All five are p-morphic images of rooted generated subframes of sameness frames: the chain by Theorem 6.1, $M_3 \cong \mathbf{F}(V_4)$, and N_5 , $1 \oplus 2^2$, $2^2 \oplus 1$ as images of $\mathbf{F}(A_4)$, $\mathbf{F}(C_{12})$, $\mathbf{F}(Q_8)$ and of ten 4-point-poset categories (`level_logic2_pmorph2.py`).*

By contrast, none of N_5 , $1 \oplus 2^2$, $2^2 \oplus 1$ is *isomorphic* to $\text{Same}(C)$ for any poset category on ≤ 5 points, any monoid of order ≤ 3 , or the groups tested (`level_logic2_survey.py`); for groups this is a theorem:

Lemma 6.3. *No group has $\text{Sub}(G) \cong N_5$, $1 \oplus 2^2$ or $2^2 \oplus 1$.*

Proof. Such G has exactly five subgroups. Suppose first that G is a p -group of order p^n . It has a subgroup of every order p^i , hence at least $n+1$ subgroups, so $n \leq 4$; if G is cyclic it has exactly $n+1$, and five forces $G \cong C_{p^4}$, a chain. If G is not cyclic, then $G/\Phi(G)$ is elementary abelian of rank $r \geq 2$ (Burnside’s basis theorem), so G has $(p^r - 1)/(p - 1) \geq p+1 \geq 3$ maximal subgroups. For $n = 2$ the subgroups are 1, G and the $p+1$ of order p , i.e. $p+3$, which is five iff $p = 2$, i.e. $G \cong V_4$ (M_3). For $n \geq 3$ the maximal subgroups have order $\geq p^2$, so together with 1, G and a subgroup of order p there are at least six. Suppose next that two distinct primes p, q divide $|G|$, and let P, Q be Sylow subgroups; $1, P, Q, G$ are four distinct subgroups. If some Sylow subgroup is not normal, its conjugates number at least $p+1 \geq 3$ (Sylow), giving at least six subgroups. So all Sylow subgroups are normal and G is their direct product; if P , say, is not of prime order, it contains a subgroup X of order p , and X and XQ are two further subgroups — six again. Hence every Sylow subgroup has prime order and G is cyclic of square-free order, whose subgroup lattice is Boolean with $2^k \neq 5$ elements. (All groups computed in this paper agree: C_{16} and V_4 have five subgroups;

C_6 has four; C_8 four; Q_8, S_3, C_3^2, C_{12} six; $C_2 \times C_4$ eight; A_4, D_4 ten; C_2^3 sixteen.) $\square$

Theorem 6.4 (six elements). *Every finite bounded poset with at most six elements is a p-morphic image of a rooted generated subframe of a finite sameness frame. Consequently Log_{fin} and S4.2Grz have the same theorems refutable on frames of size ≤ 6 , and on all finite chains; any formula separating the two logics is refuted only on frames of size ≥ 7 .*

Proof. Sizes ≤ 5 : Proposition 6.2. Size 6: a bounded poset with six elements is $\perp + P + \top$ for a poset P on four points; there are sixteen, fifteen of them lattices and one not (the butterfly $\perp < 0, 1 < 2, 3 < \top$, Figure 4). Over 36 source categories (all posets on ≤ 4 points; $C_6, C_8, C_{12}, V_4, S_3, Q_8, A_4, D_4, C_2 \times C_4, C_3 \times C_3, C_2^3$; the parallel pair), with roots pre-filtered by the necessary conditions $\text{height } R[W] \geq \text{height } T$ and $\text{width } R[W] \geq \text{width } T$ (which follow from $f(R[x]) = S[f(x)]$ for a p-morphism f), an explicit p-morphism was found for fifteen of the sixteen, including the butterfly (from a 4-point-poset category with $|\text{Same}| = 8$); each is a finite certificate checked by machine and listed explicitly in the supplementary file `certificates.json` (`level_logic3_cert.py`). The sixteenth is the 6-chain, which is a sameness frame by Theorem 6.1. The consequence for the logics is preservation of validity under p-morphic images and generated subframes. $\square$

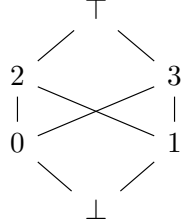


FIGURE 4. The butterfly: a bounded poset that is not a lattice ($0 \vee 1$ does not exist), yet a p-morphic image of a sameness frame — lattice sources have non-lattice images, just as modular sources have non-modular ones (Theorem 4.5).

Alt text: Hasse diagram of a six-element poset: a bottom element; above it two incomparable elements 0 and 1; above them two incomparable elements 2 and 3, each of 0 and 1 lying below each of 2 and 3; and a top element above 2 and 3.

Remark 6.5 (honest boundary). Theorem 6.4 does not settle $\text{Log}_{\text{fin}} = \text{S4.2Grz}$. Size ≥ 7 needs taller and wider sources (e.g. 5-point-poset categories) and a p-morphism search better than backtracking for negative

instances; and the iso-realization question of Lemma 6.3 shows that the sameness frames are a proper subclass of finite lattices. Whether some finite bounded poset is *not* a p-morphic image of any finite sameness frame — which would make Log_{fin} strictly stronger than S4.2Grz — is the main open question of this paper. The exact identity of Log_{all} , which lies between S4.2 and Log_{fin} , is a second one.

7. COMPUTATIONAL RECORD

All finite claims are verified by exhaustive computation in `scripts/comparison/`: `level_logic.py` (§3), `level_logic2.py` (Theorem 4.1, Table 1), `level_logic2_survey.py` and `level_logic2_pmorph2.py` (Proposition 6.2, Lemma 6.3), `level_logic4_modular.py` (Theorems 4.3, 4.5; certificate `cert_modularity.json`), `oo_completeness.py` (Theorem 5.1), `level_logic3_cert.py` (Theorem 6.4; the eighteen computed p-morphisms for the five- and six-element targets are listed in `certificates.json`). All certificates and scripts are supplied as supplementary material.

8. CONCLUSION

A category has a modal logic, and it is informative: it contains S4.2 always, reaches S4.3 exactly on chains — for finite groups, exactly on cyclic p -groups — and reads every finite bound on the width and height of $\text{Same}(C)$ off the axioms bw_n, bd_n , while being provably blind to distributivity and modularity, with both blind spots witnessed inside the class by group frames. Its intended models are quotients that collapse along accessibility, a counterpart-style modal setting with monotone contingent identity whose identity/encapsulation fragment is complete for abstract models. The logic of finite sameness frames agrees with S4.2Grz through size six and on all finite chains, while infinite frames drop Grz . Two problems remain: the realization problem — which lattices, and which families of equivalences, come from a category — which governs completeness for the intended models; and the exact identity of the level-shift logic, which the same problem controls through p-morphic images.

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Availability of data and materials. All verification scripts and p-morphism certificates named in §7 are provided as supplementary material with the submission.

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